

From Physics to Quantum Machine Learning

Internship at INMA — Instituto de Nanociencia y Materiales de Aragón

Exploring quantization strategies for neural-network ansätze in variational quantum Monte Carlo.



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1. Internship context

Where & with whom

- INMA, at the Facultad de Ciencias, University of Zaragoza.



- Supervisors: Jesús Carrete Montaña & David Zueco Láinez
- 15 June – 20 July 2026

What I was asked to investigate

- Can lower numerical precision reduce computational cost?
- Can we preserve the accuracy and physical properties of neural-network quantum states?
- Start simple, validate against some reference data, then move to a more advanced architecture.

2. Why this problem?



**Quantum many-body
problems are expensive**

**Neural networks can
approximate quantum states**

**Lower precision may make
computation cheaper**

- The project combines three areas I have encountered during my master's: physics and numerical simulations.
- The key trade-off was accuracy versus computational cost.
- For physics, accuracy means more than a good energy: the quantum state should also retain its physical properties.

3. The basic idea: quantization

- A standard neural network uses relatively high numerical precision (FP32).
- Quantization represents parameters/computations with fewer bits: FP16, INT8, INT4, INT2.
- Fewer bits can reduce memory use and computation but too much reduction can damage the model.
- I tested two strategies: PTQ (after training) and QAT (quantization included during training).

**FP32
baseline**

FP16

INT8

INT4

INT2

**Question: how far can I reduce precision
before physics is lost?**

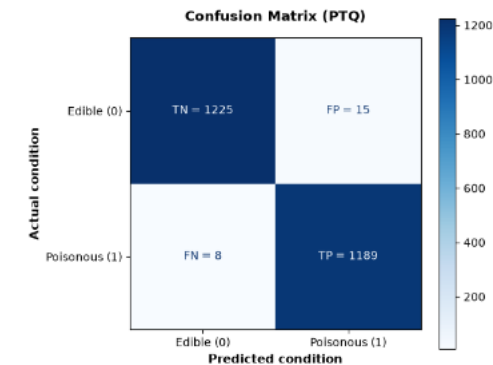
4. First exercise: a simple neural network

Mushroom dataset.

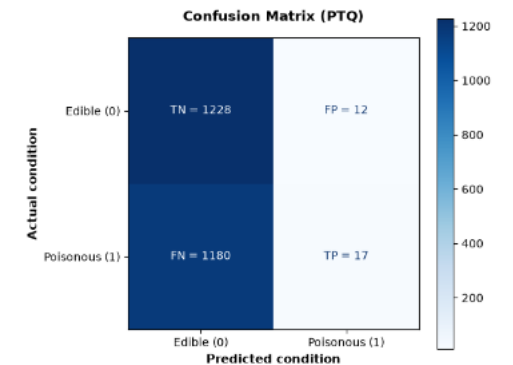
- A simple classification problem gave me a controlled environment.
- It let me learn JAX, Flax, Optax and Qwix before working with quantum states.
- I trained an MLP on the UCI Mushroom dataset.

What I tested

- FP32 baseline
- PTQ: weight-only and dynamic-range quantization
- QAT: quantization-aware training
- Accuracy, latency and confusion matrices



(a) Confusion matrix for weight only quantization with arch = [64, 2], epochs = 100 and INT2.



(b) Confusion matrix for weight only quantization with arch = [4, 2], epochs = 50 and INT2.

Figure 8: Confusion matrices for the weight only quantification model using data type: INT2.

5. First exercise results

- The measurements below belong to the internship's logbook:

Table 2: PTQ performance comparison for arch = [4, 2] and epochs = 50.

Data Type	Accuracy (%)	Accuracy Loss (%)	Latency (ms)	Speed up
FP32	87.32	0.00	0.046774	1.00x
INT8	87.32	0.00	0.045655	1.02x
INT4	87.24	0.08	0.043030	1.09x
INT2	51.09	36.23	0.045569	1.03x

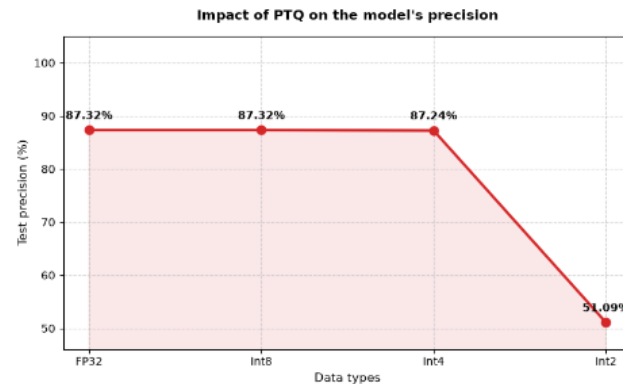


Figure 6: Impact of PTQ on the model's accuracy for arch = [4, 2] and epochs = 50.

Table 5: QAT performance comparison for arch = [64, 2] and epochs = 100.

Model	Accuracy (%)	Latency (ms)
FP32	99.63	0.066366
QAT INT2	99.30	0.056841
PTQ INT2	99.06	0.063868

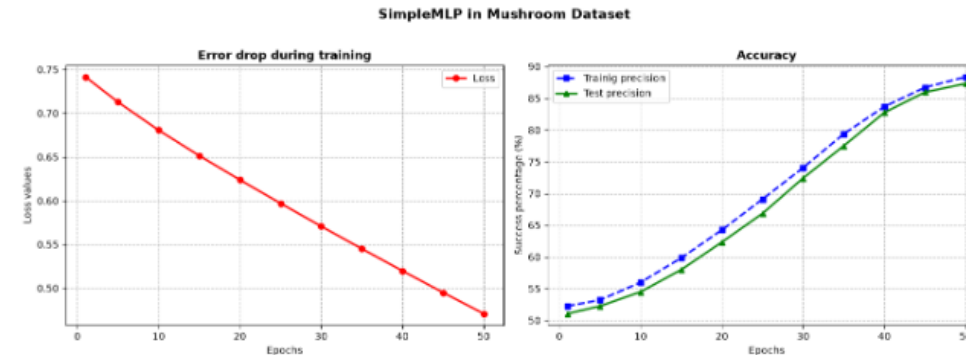
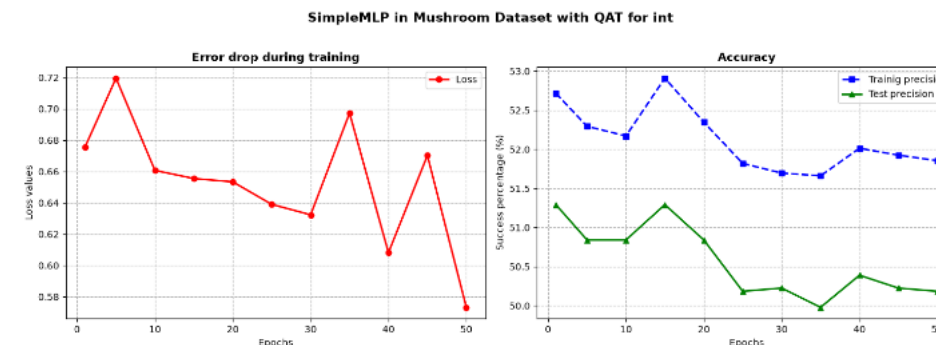


Figure 4: Simple MPL model for arch = [16, 2] and epochs = 50.



(a) Simple MPL model for arch = [4, 2] and epochs = 50.

6. Moving to physics: the transverse-field Ising model



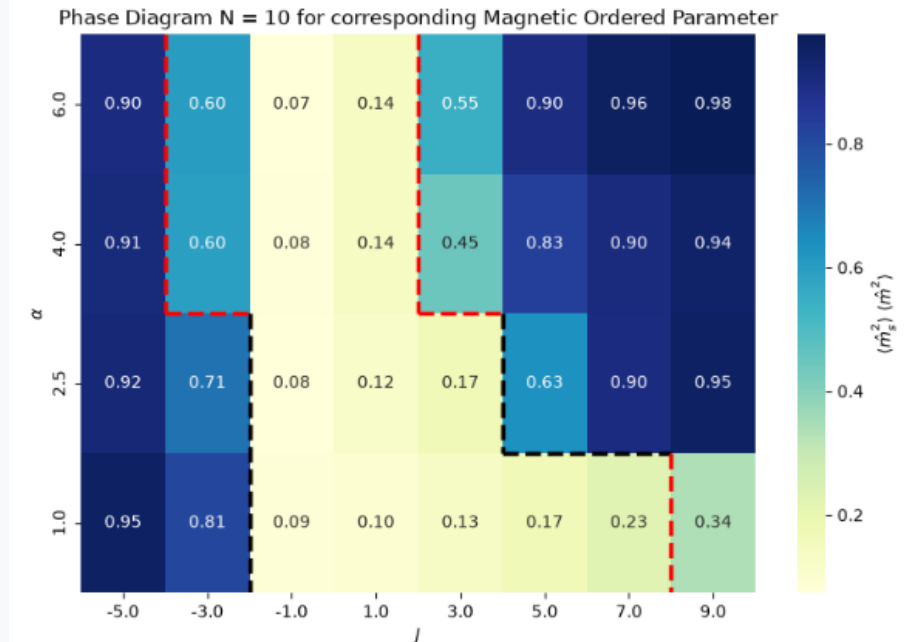
- I then moved from ordinary classification to a quantum many-body problem.
- The 1D transverse-field Ising model (TFIM):

Hamiltonian

$$H = \Gamma \sum_i \sigma^x_i + V \sum_i \sigma^z_i \sigma^z_{i+1}$$

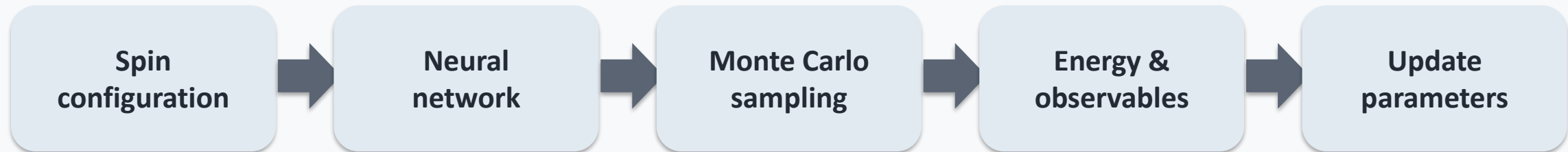
- For the system sizes I used, exact diagonalization was still possible and gave me a ground-truth reference.

Goal: learn the ground-state wavefunction



(a) Phase diagram measuring $\langle \hat{m}_s^2 \rangle$ and $\langle \hat{m}^2 \rangle$.

7. How the VQMC calculation works



- The neural network represents the logarithm of the wavefunction amplitude.
- NetKet uses Metropolis sampling rather than explicitly summing the whole Hilbert space.
- The energy is minimized, while overlap and spin correlations provide additional checks.

8. First TFIM results:



Exact diagonalization

$$E = -25.491 \text{ J}$$

FP32 neural network

$$E \approx -25.428 \text{ J}$$

Overlap = 0.996

Interpretation

- The FP32 network reproduced the ground state very well.
- This gave me a validated baseline before changing numerical precision.

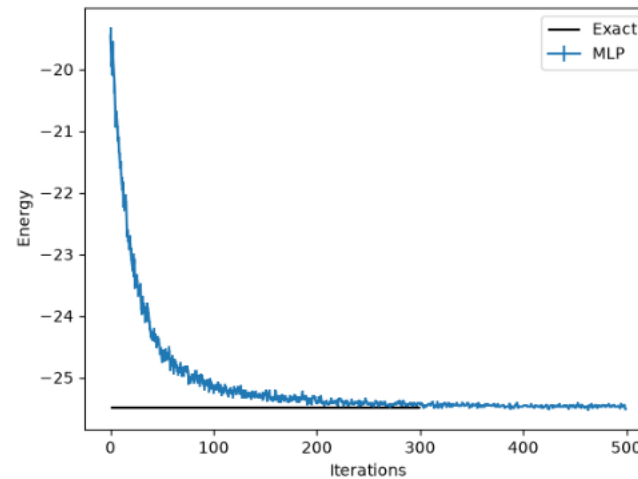


Figure 14: Energy convergence comparison between MLP and ED.

9. PTQ and QAT on the TFIM:

- The measurements below belong to the internship's logbook:

Table 8: PTQ performance comparison for the 1D TFIM using different data types.

Data Type	Energy (H)	Relative Error	Correlator	Corr Error	Overlap	Latency (ms)	Speed Up
FP32	-25.427646	0.002485	10.818816	0.033432	0.996269	0.602828	1.00x
FP16	-25.472749	0.0007155	10.765304	0.086945	1.000504	0.336029	1.79x
INT8	-25.464798	0.001027	10.726541	0.125708	0.997199	0.234925	2.57x
INT4	-25.372614	0.004644	11.746746	0.894497	0.958133	0.305106	1.98x
INT2	-22.182162	0.129804	16.589086	5.736837	0.200938	0.665848	0.91x

Table 9: QAT performance comparison for the 1D TFIM using different data types.

Data Type	Energy (H)	Relative Error	Correlator	Corr Error	Overlap	Latency (ms)	Speed Up
FP32	-12.781754	0.000246	5.425310	0.012575	0.990462	0.574493	1.00x
FP16	-12.766040	0.001476	5.389978	0.019006	0.965745	0.193748	2.97x
INT4	-12.758444	0.002070	5.190808	0.055255	0.995119	0.359624	1.60x

Where the results for the exact diagonalization were:

Exact energy (ED): -12.784906 J .

Exact correlator (ED): 5.494405.

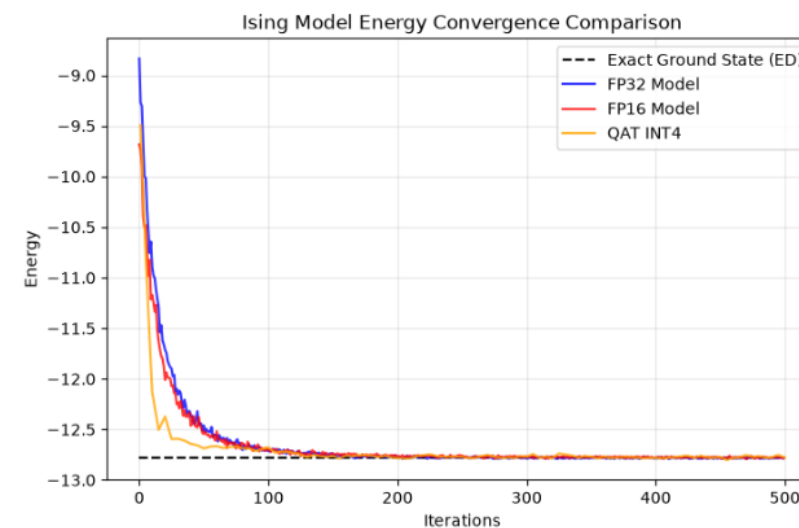


Figure 16: Energy convergence comparison between models.

10. A problem appeared: symmetry

- When I checked overlap across the ferromagnetic phase, the quantized models gave unexpectedly poor values.
- The TFIM has a Z_2 spin-flip symmetry: flipping all spins leaves the energy unchanged.
- I modified the architecture so $f(x)$ and $f(-x)$ are explicitly combined.

Before

$f(x)$

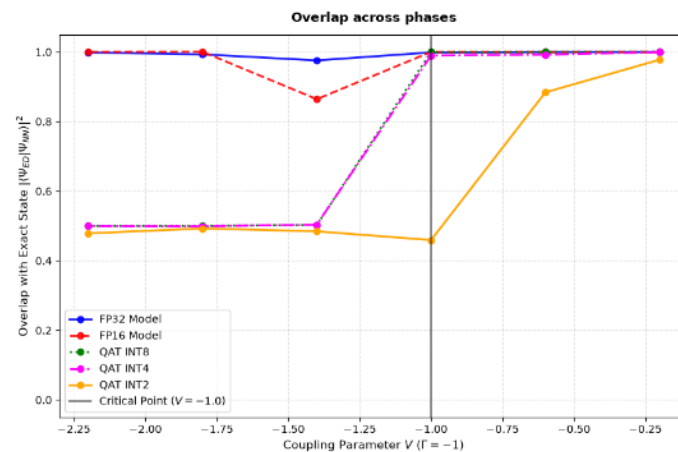


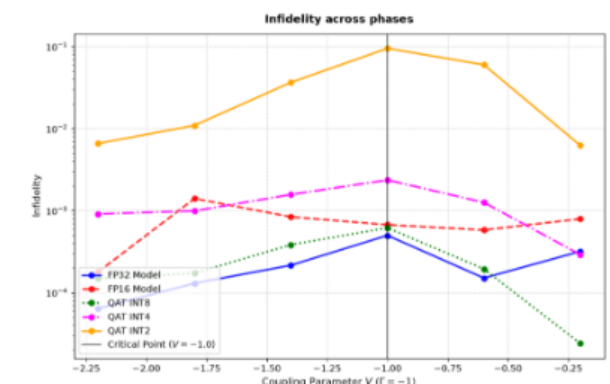
Figure 18: Overlap for different data types across the ferromagnetic phase.

After

$\frac{1}{2}[f(x)+f(-x)]$



(a) Overlap for different data types across the ferromagnetic phase.



(b) Infidelity error for the different overlap measurements.

Figure 21: Overlap for different data types across the ferromagnetic phase when taking into account Z_2 symmetry.

11. From MLP to Vision Transformer (ViT)

Why change the architecture?

- The [reference project](#) used a Vision Transformer (ViT).
- I wanted to test whether the quantization strategy also worked for a more advanced ansatz.
- This was also the point where implementation became substantially more challenging.

What I did

- Reused the TFIM Hamiltonian, sampler and Z_2 symmetry idea.
- Reused the [working code](#) for the ViT architecture from the reference paper.
- Implemented $\text{FP32} \rightarrow \text{FP16} \rightarrow \text{PTQ} \rightarrow \text{QAT}$.
- Compared energy, overlap, V-score and latency.

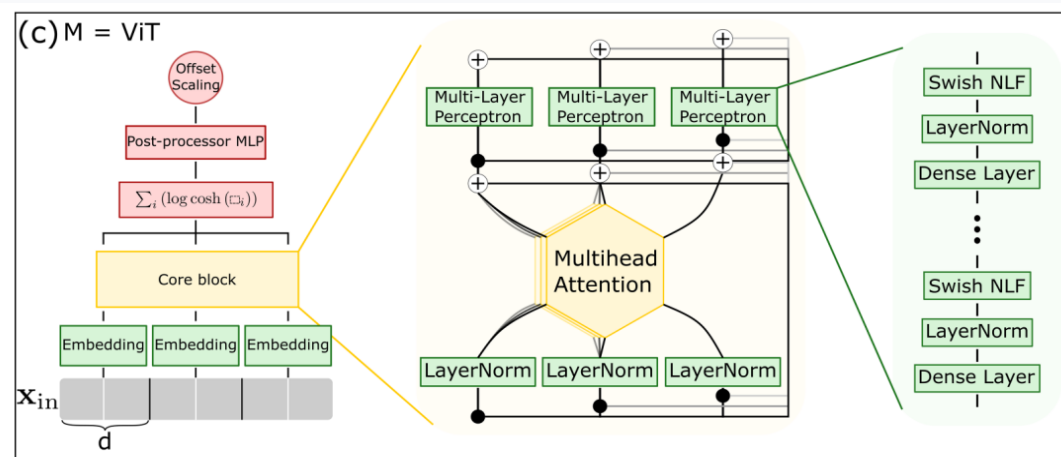


Figure describing the structure of the ViT used (image taken from [reference paper](#)).

12. Main results: ViT + quantization

- The measurements below belong to the internship's logbook:

Exact energy (ED): $-12.784906 J$.

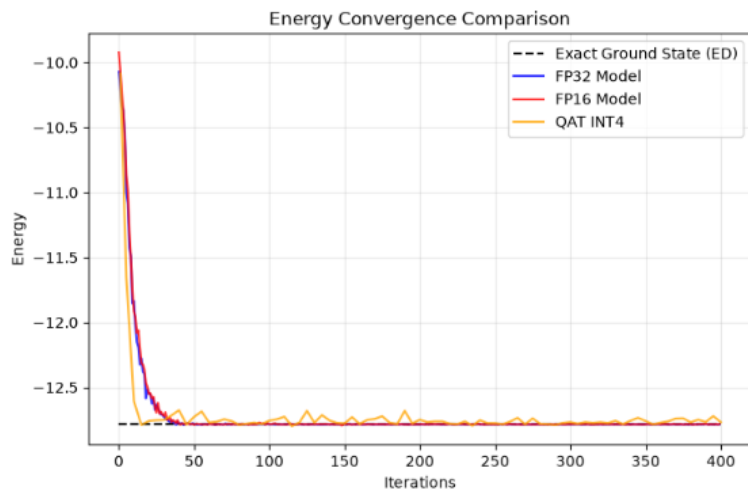


Figure 24: Energy convergence comparison between models using the ViT for QAT.

Table 11: PTQ performance comparison for the 1D TFIM (ViT) with $\mathbb{Z}_2$ symmetry using different data types.

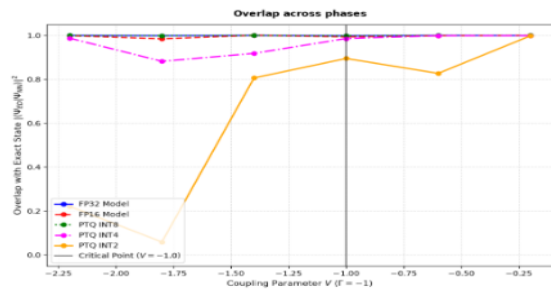
Data Type	Energy (H)	Relative Error	Overlap	V-score	Latency (ms)	Speed Up
FP32	-12.788336	0.000268	0.998721	0.000333	0.657260	1.00x
FP16	-12.790295	0.000421	0.996735	0.000465	0.347614	1.89x
INT8	-12.784411	3.875470e-05	0.998784	0.000047	0.541015	1.21x
INT4	-12.508474	0.0216218	0.944502	0.007123	0.648870	1.01x
INT2	-12.283744	0.039200	0.901989	0.020836	0.731366	0.90x

Table 12: QAT performance comparison for the 1D TFIM (ViT) with $\mathbb{Z}_2$ symmetry using different data types.

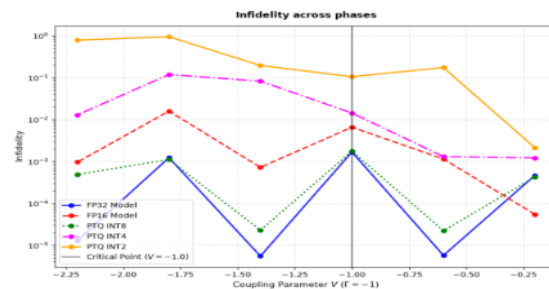
Data Type	Energy (H)	Relative Error	Overlap	V-score	Latency (ms)	Speed Up
FP32	-12.784949	3.329011e-06	0.999998	0.000101	2.736734	1.00x
FP16	-12.785411	3.949304e-05	1.000039	0.000040	0.316519	8.65x
INT4	-12.783785	8.770330e-05	0.992943	0.000071	0.778351	3.52x

12.1. Main results: ViT + quantization

- The measurements below belong to the internship's logbook:

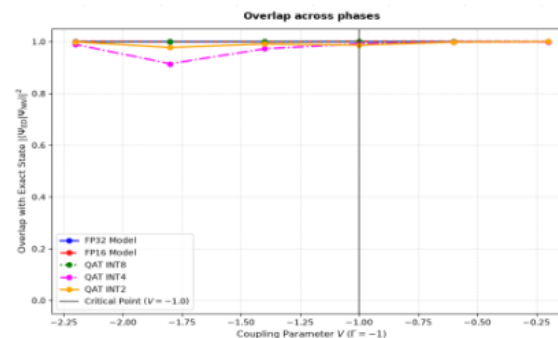


(a) Overlap for different data types across the ferromagnetic phase.

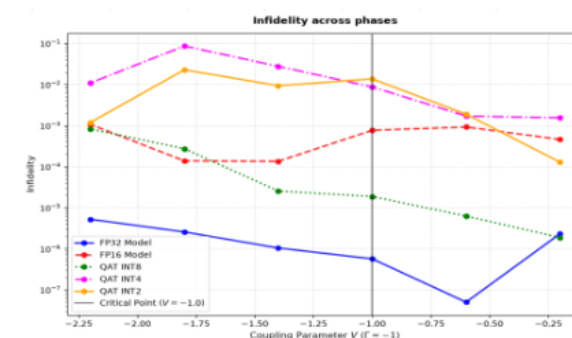


(b) Infidelity error for the different overlap measurements.

Figure 23: Overlap for the PTQ implementation of the TFIM using the ViT across the ferromagnetic phase when taking into account $\mathbb{Z}_2$ symmetry.



(a) Overlap for different data types across the ferromagnetic phase.



(b) Infidelity error for the different overlap measurements.

Figure 26: Overlap for the QAT implementation of the TFIM using the ViT across the ferromagnetic phase when taking into account $\mathbb{Z}_2$ symmetry.

13. What did I actually learn as a physicist?

Physics preparation

- Quantum mechanics helped me understand the Hamiltonian, symmetries and observables.
- Statistical/computational physics prepared me for Monte Carlo and numerical experimentation.
- Physics trained me to compare models against a reference and question results.

What I needed to learn on the job

- Modern ML frameworks: JAX, Flax, NetKet, Qwix.
- Working with GPUs and uv environments.
- Reading someone else's code and turning a paper into an implementation.

14. Reflection: opportunities and limits of physics

- Physics gave me the concepts needed to decide what should be measured and what counts as a meaningful result.
- But the internship also showed me the limits of purely physics-based intuition: implementation details, software tools and ML training behaviour can dominate the outcome.
- The most useful skill was moving back and forth between theory, code and numerical evidence.
- I became more comfortable working independently while still using supervision to check the direction of the project.

Theory → implementation → validation → reflection

15. What I would improve / what comes next

What I would have liked more training in

- Practical ML / deep-learning courses during my master's programme.
- GPU computing.

Possible continuation

- Test larger system sizes and more parameter regimes.
- Investigate more extreme quantization ([BinaryAttention](#)).

16. Take-home message

1

I learned to work from an existing research codebase to an independently tested implementation.

2

Quantization can reduce computational cost while preserving useful physics but only within the right precision and training regime.

3

The internship taught me that being a physicist in computational research means connecting theory, numerical methods, software and critical evaluation.

Thank you — questions?